the vehicle can negotiate the curve safely. It can be seen from figure 30 that a ground vehicle (tire pressure of 25 pounds per square inch) entering a curve at ground speeds in excess of the corresponding tire hydroplaning speed (52 miles per hour) could not negotiate road curves at posted speed limits under pavement conditions conducive to hydroplaning since the friction coefficient for hydroplaning tires (μ does not exceed 0.05) produces vehicle slide-out speeds less than the posted speed limit speeds.

Low Friction Coefficients Not Associated With Tire Hydroplaning

The large mass of data just described suggests that tire hydroplaning can be reasonably explained in terms of fluid density effects alone and equation (4), derived on this basis, is seen to give good estimates of tire hydroplaning speed values. (See fig. 4.) Some data exist, however, that show a complete loss of tire braking traction (one manifestation of hydroplaning) occurring at ground speeds considerably less than the tire hydroplaning speed.

Such a loss at these lower speeds cannot be ascribed to hydroplaning from fluid density effects since the fluid dynamic pressures developed at these lower speeds is insufficient to lift the tire off the pavement surface. Since this type of braking traction loss occurs only when smooth tires are used on smooth wet pavement surfaces or when rib tread tires are used on very smooth wet pavement surfaces, it probably arises from thin-film lubrication effects on the tire-ground surfaces in which fluid viscous properties, previously ignored, tend to predominate. Two examples of thin-film lubrication are given.

The first example is shown in figure 31 which indicates the loss in braking traction experienced by a smooth tread aircraft tire when braked on a relatively smooth wet concrete runway at tire pressures of 120 and 260 pounds per square inch. It can be seen in part (a) of this figure that at ground speeds of 95 to 100 knots, both the $p = 260$ pounds per square inch curve and the $p = 120$ pounds per square inch curve drop to the residual free-roll friction coefficient. This result indicates complete loss of braking traction at this speed and is one of the manifestations of total tire hydroplaning. When these data are plotted against the velocity ratio V_G/V_P as in part (b) of this figure, it can be seen that the curve for $p = 260$ pounds per square inch reaches the free-rolling friction coefficient level (zero braking traction) at a ground speed 35 percent less than the total hydroplaning speed required by equation (4).

A similar result is shown in figure 32 for a rib tread aircraft tire when braked on a very smooth wet membrane that was placed on top of the track runway. It is of interest to note that the water depth for this test was insufficient to form puddles on the membrane. The ground speed for this particular test was 86 knots which gives a value of velocity ratio V_G/V_P of 0.68 and an estimated hydroplaning speed of 127.5 knots. Since the average friction coefficient developed for this speed is about 0.05 (approximately equal to tire free-rolling resistance), the loss in braking traction must be nearly complete at a ground velocity at least 32 percent less than the hydroplaning speed predicted by equation (4).